

1. Problem

Official documents are falsified through edited text, replaced photos, forged stamps and altered identifiers — and now through AI generation. Manual review is slow, repetitive, and misses precisely the subtle cases. The reviewer rarely lacks judgement: they lack time, and a place to look.

2. Target users

User The person who checks documents at volume: a credit analyst, an HR recruiter, a compliance officer, an administrative clerk. **Buyer** The head of risk, compliance or operations, who buys reduced exposure rather than saved minutes.

3. Solution — screening, not certification

What the product claims, and what it refuses to claim

DocShield AI does not certify documents. It surfaces suspicious signals and the evidence behind them, so a human reviewer decides faster and better: every flag is drawn and numbered on the document itself. A document that is fabricated but internally coherent passes every check here — screening detects inconsistency, not falsity, and authenticity can only be established with the issuer.

Handled by the model

- ✓ **Reads** the document and extracts fields.
- ✓ **Sees** visual manipulation: mixed fonts, broken alignment, re-inserted text.
- ✓ **Compares** pages against one another.

Never trusted to the model

- ✗ **Counting digits** — identifier length.
- ✗ **Checksums** — IBAN mod-97 key.
- ✗ **Arithmetic and calendars** — VAT, totals, impossible dates, inverted periods.

Deterministic code: exact and auditable.

4. Prototype — deployed and open

Upload → Normalization → Multimodal analysis → Deterministic checks → Risk score → Report

PDF, JPG and PNG. Every page is rendered and sent together, so a value changing between two pages is visible to the model. Preview and model input share the same images, so a flagged region lands where the reviewer looks. Output passes a strict JSON schema, re-validated server-side; a failed deterministic check raises the score floor and blocks an “accept” verdict.

5. Validation — the loop that shaped the build

Build → Test → Failure → Diagnose → Fix → Retest → Learn

Five reviewers from five companies: four ran documents through the deployed prototype, one assessed the product without testing it. The first reported four defects, each traced to a specific line before anything was changed, all four corrected within the challenge window. Deliverable 03 records every message verbatim, failures included.

6. Key learnings — and what each one is worth

PROVEN	A planted date inconsistency detected, named and localised on a document designed by a third party. Genuine certified documents passed unflagged.
FIXED	Four reported defects corrected and deployed; two independently observed working afterwards.
TO RETEST	Cross-page fix deployed and pipeline verified, but the two-page invoice that exposed the gap has not been re-run — end-to-end detection not yet demonstrated.
LIMIT	A well-made or AI-generated document that is internally coherent is missed by design.

7. Business logic

Value Less manual screening, faster triage, a traceable record of every check. Minutes saved and the cost of a missed forgery are *not* measured yet, and no figure is asserted here.

Model (hypothesis) Per document, per seat, or an integration API — to be tested in pricing conversations, not assumed.

Beachhead (hypothesis) Lending and leasing in Morocco, for volume and direct loss per miss. Two reviewers pointed instead to public administration and education; the question is open.

8. Next step

Reliability before reach. Measure false positives and false negatives on a controlled document set, re-run the outstanding test, then pilot with one organization before adding any new detection category.

Key assumption. Organizations will adopt AI-assisted screening only if every flag carries transparent evidence. An unjustified score will not be trusted — and should not be.